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genCell: a diffusion-based generative model for conditional single-cell RNA sequencing data generation

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1.Introduction

- Literature Review
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Single-cell RNA Sequencing technology

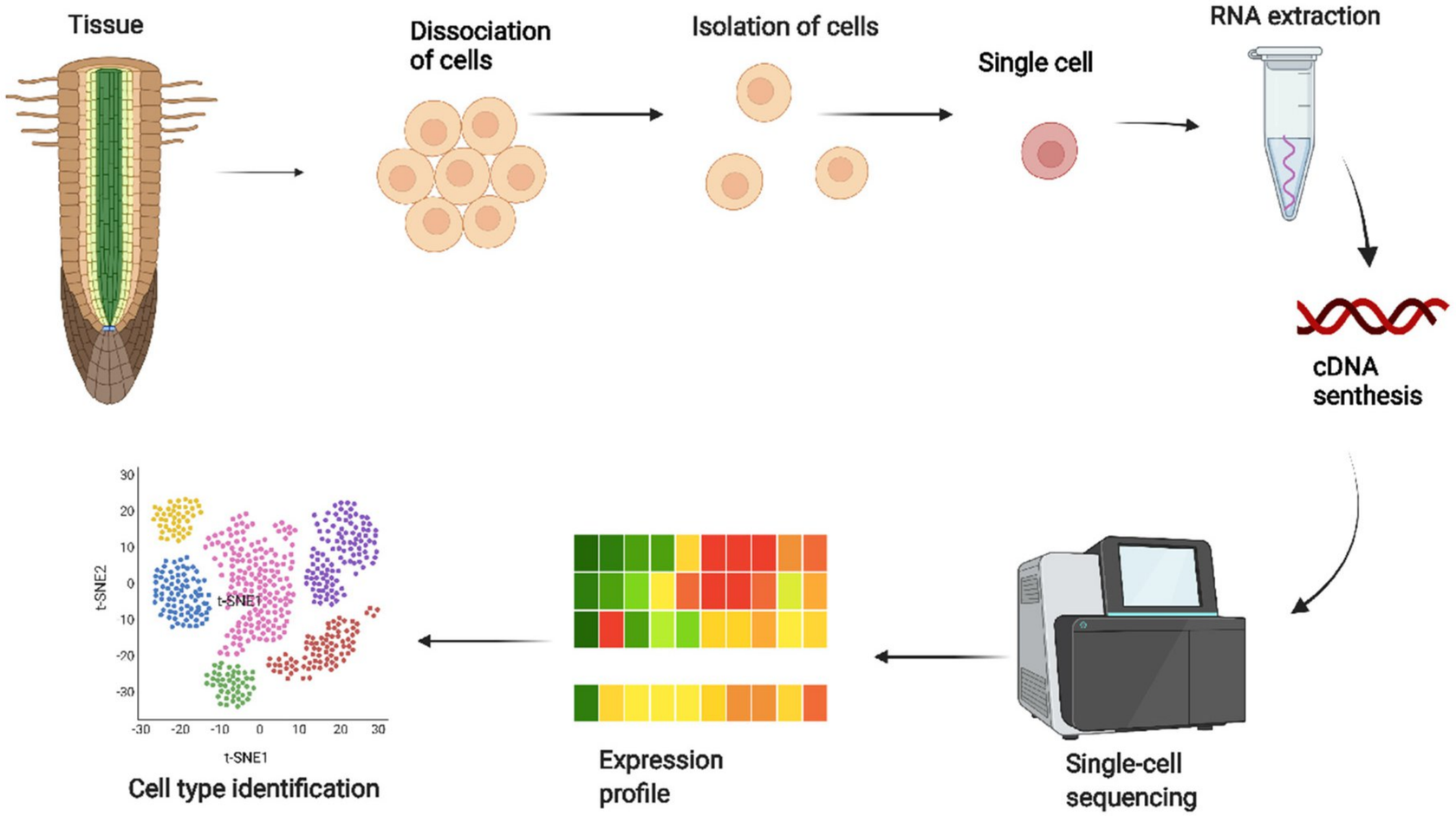

Figure 1. Workflow for single-cell RNA sequencing, allow us to do analysis of individual cells [1]

Generative AI in single-cell - Downstream tasks

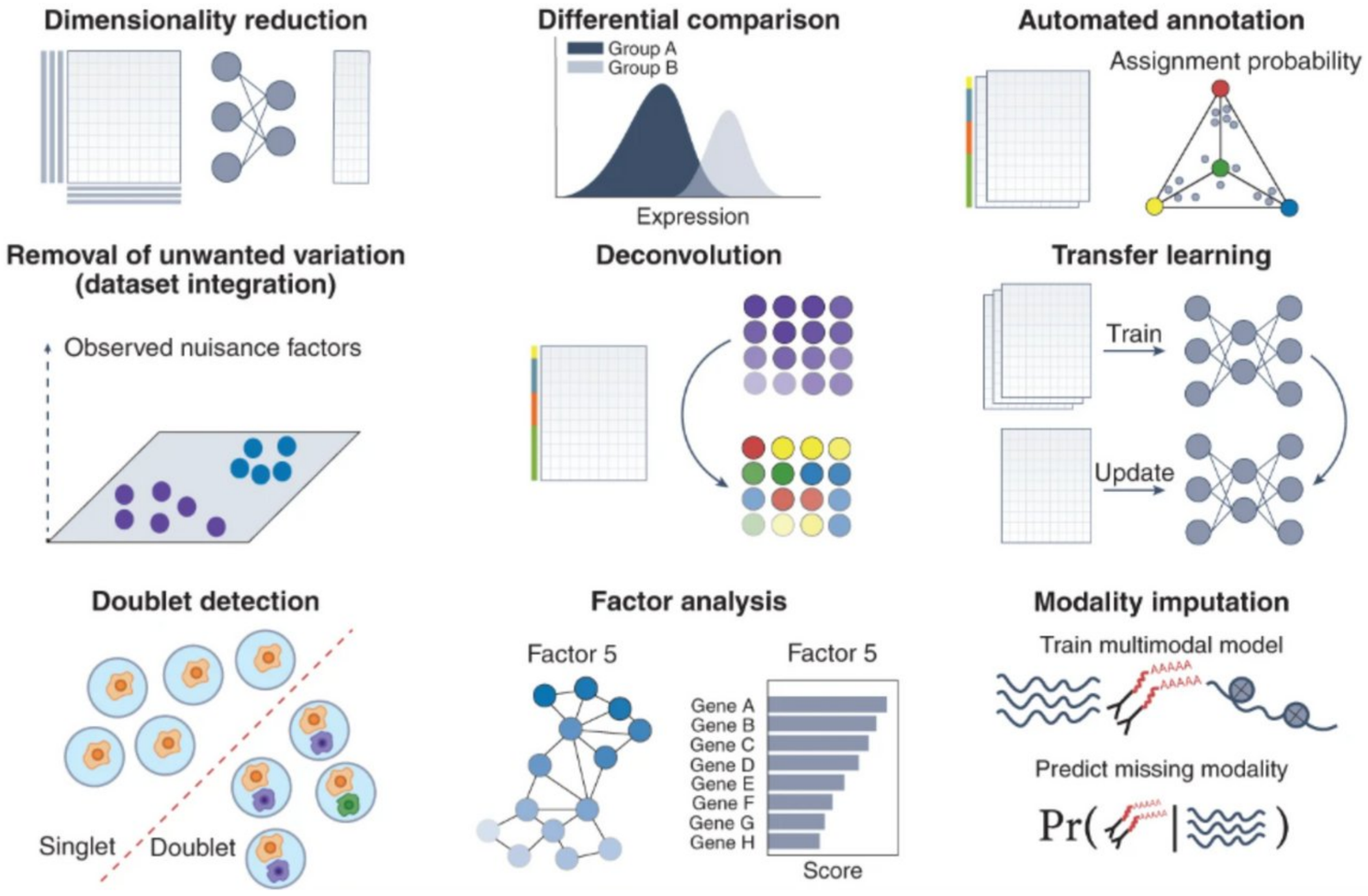

Figure 2. scVI: a generative VAEs based model using for downstream tasks [2]

Generative AI in single-cell - Downstream tasks

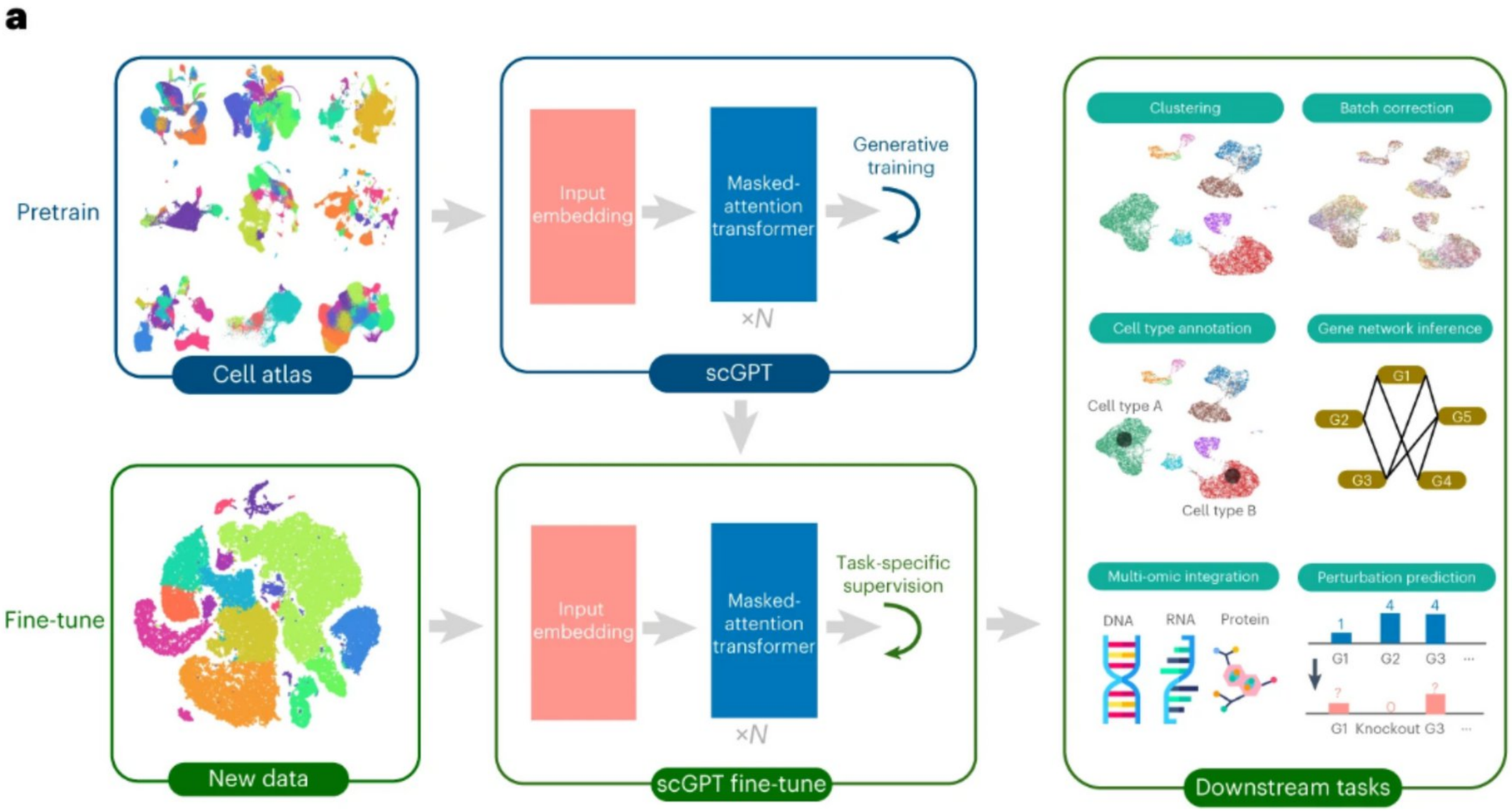

Figure 3. scGPT: a generative GPT based foundation model for downstream tasks [3]

Research Gaps

- **Limited high-quality scRNA-seq data** [4]
  - Scarce samples at key development disease stages
  - High cost, privacy restrictions and batch effect

- **Lacks of generative AI model applied for single-cell data generation**
  - scVI (VAE-based) and scGPT only used for downstream tasks (clustering, batch effect correction or dimensionality reduction)

- **Diffusion model** [5] performed excellent in image/language fields but remains **under-explored** in Bioinformatics research (single-cell analysis)

Objectives

- Building genCell, a diffusion-based generative model to generate high-resolution scRNA-seq data under biological conditions
- Adopting the model to generate scRNA-seq data of Lung Fibrosis disease
- Studying the effects of prior distributions in model training behavior and performance.

2.Material & Methodology

- Data and framework
- Pipeline
- Model architecture
- Evaluation Metrics

Data and Framework

- **Data source:** Tabular Muris [6]
- **Data description:**
  - 45602 sample of 10 different cell types
  - 128 features (gene expression)
- **Framework:** scanpy (data prerocessing, visualization), PyTorch (model implementation), seaborn/matplotlib (visualization)
- **Data preparation:** the data is divided into 80-20% for train and evaluate sets.

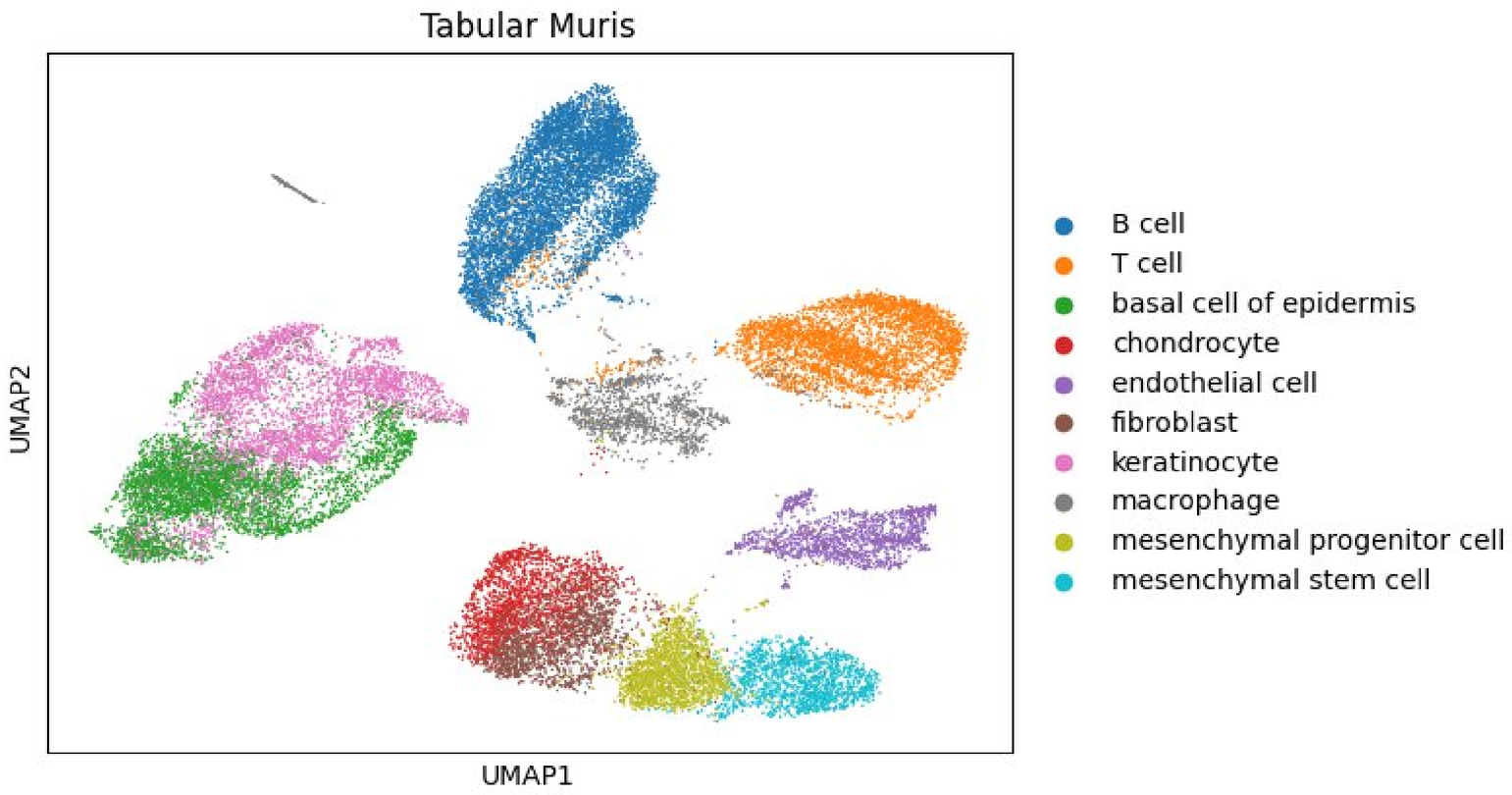

Figure 4. Tabular Muris. UMAP visualization for 10 most common cell types

Pipeline

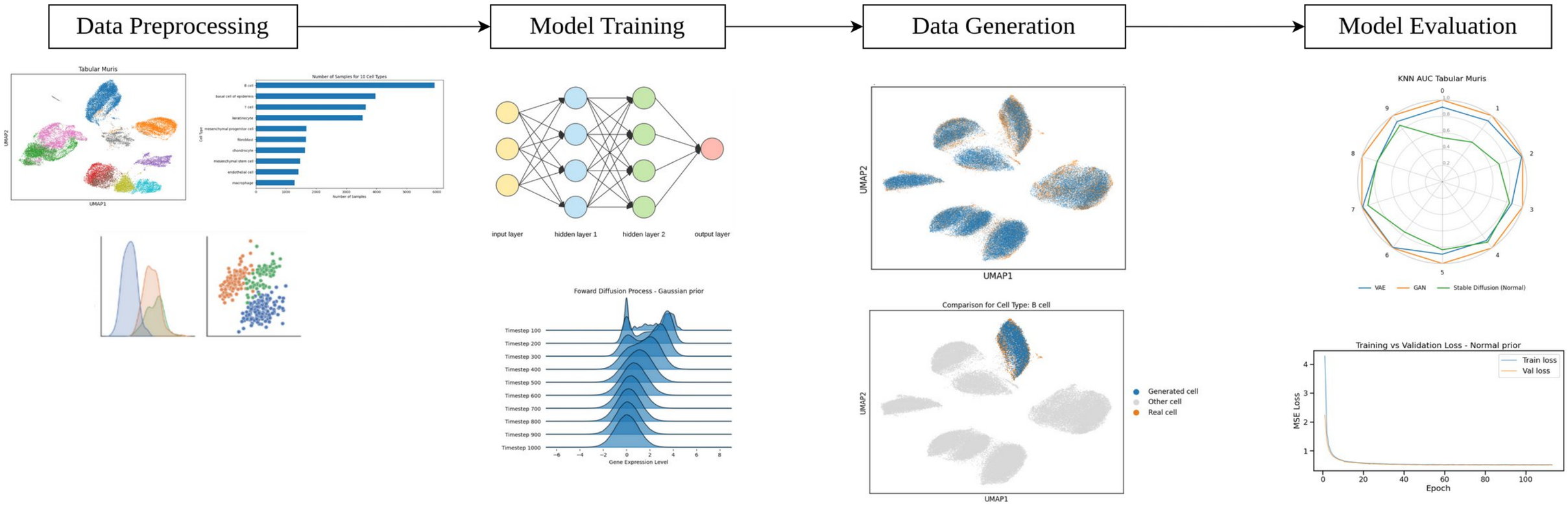

Figure 5. Project overall pipeline. Consists of 4 main stages:
Data preprocessing, Model Training, Data Generation and Model Evaluation

Model architecture

Architecture descriptions.

- Forward diffusion
  - Adds noise to gene expression matrices
  - $X_0 \rightarrow X_T = X_0 + \epsilon$
- Reversed diffusion
  - Predicts noise ϵ_θ using deep neural network blocks.
  - Removes predicted noise to generate new expression matrix X^*_0

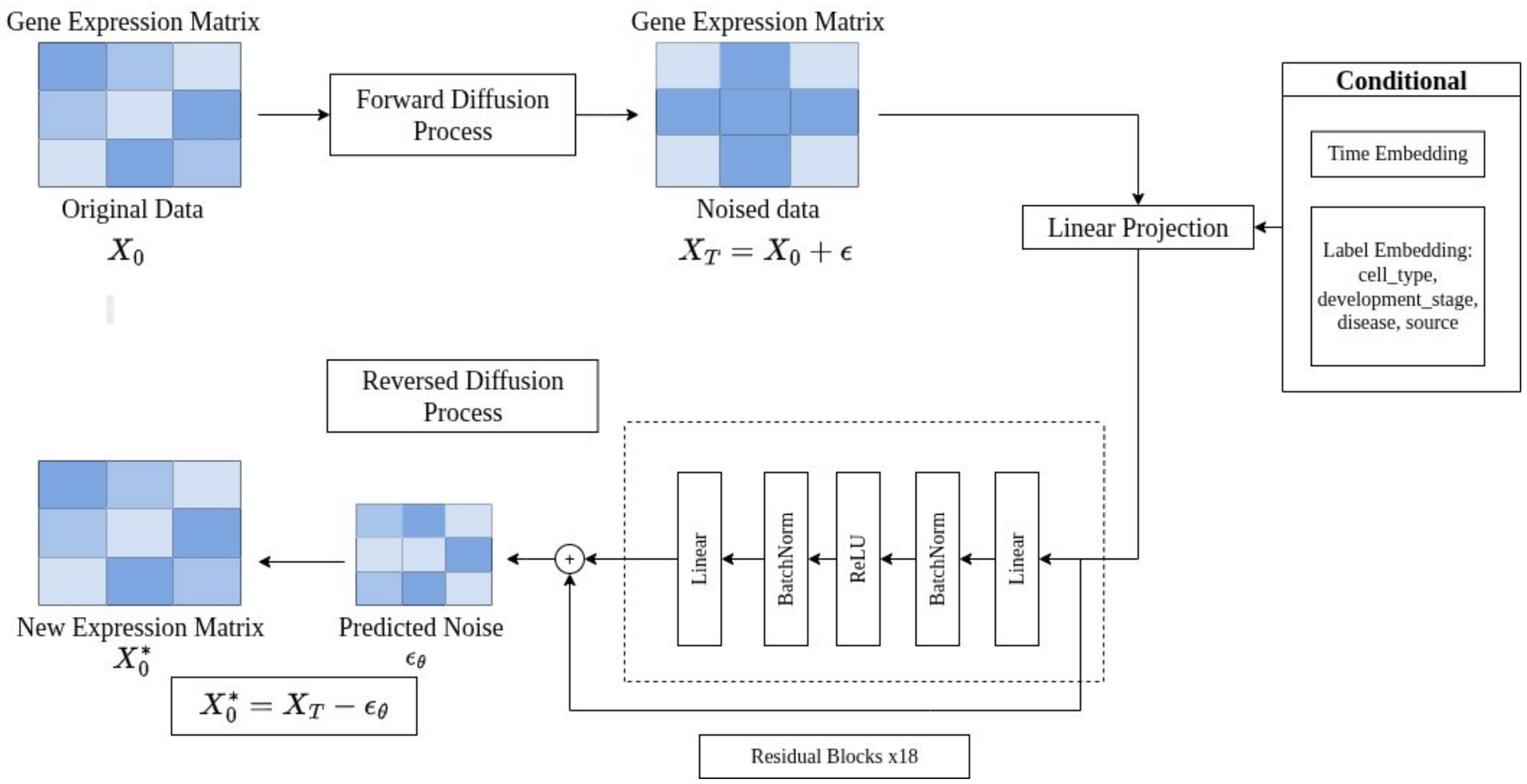

Figure 6. genCell model architecture.
Containing 2 main processes: Forward and Reversed Diffusion.

Evaluation Metrics

- **Statistical distance metrics:** Energy distance (ED), Maximum Mean Discrepancy (MMD), Spearman's Rank Correlation Coefficient (Spearman SCC).
  - **ED, MMD:** measuring the similarity between the data distribution of real data and generated data
  - **SCC:** measure the correlation (consistently increasing or decreasing) between sample (cell) and feature (gene)

- **Non-statistical metrics**: UMAP visualization, KNN-AUC score.
  - **UMAP:** compare the plot of real data and generated data.
  - **KNN-AUC score:** measure the ability of a KNN model to distinguished between real data and generated data (≈ 0.5 means the data is indistinguishable)

3.Experiment & Result

- Statistical Metrics
- Non-statistical Metrics

Statistical Metrics

- **Diffusion models (genCell) outperform VAEs and GANs** in terms of conditional scRNA-seq data generation
- The choice of prior can significantly influence model performance and output quality

Models	Energy Distance	MMD	SCC
Diffusion - Normal prior	**0.256**	**0.004**	**0.996**
Diffusion - Student's t prior	**0.255**	**0.005**	**0.995**
Diffusion - Log normal prior	6.45	0.511	0.837
VAEs	1.914	0.139	0.995
GANs	1.814	0.219	0.910

Table 1. Model performance comparison. **Bold**: best, Red: worst

Non-statistical Metrics

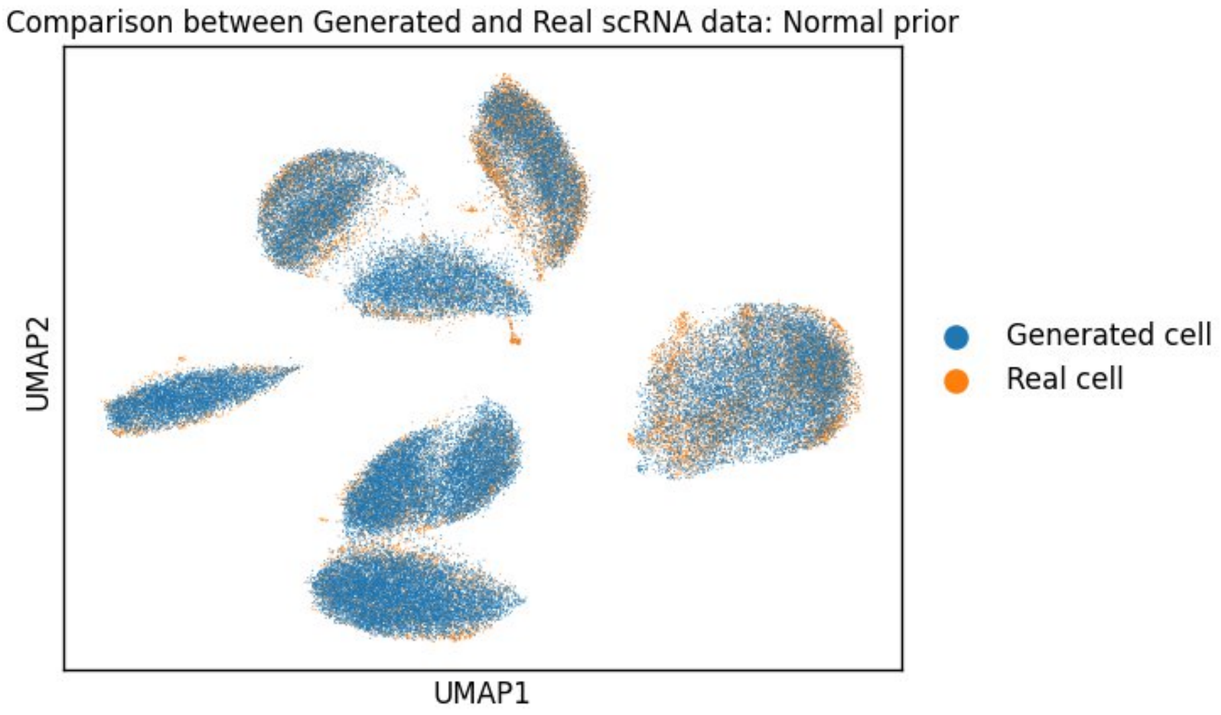

Figure 7. UMAP of real data and generated data (Normal prior)

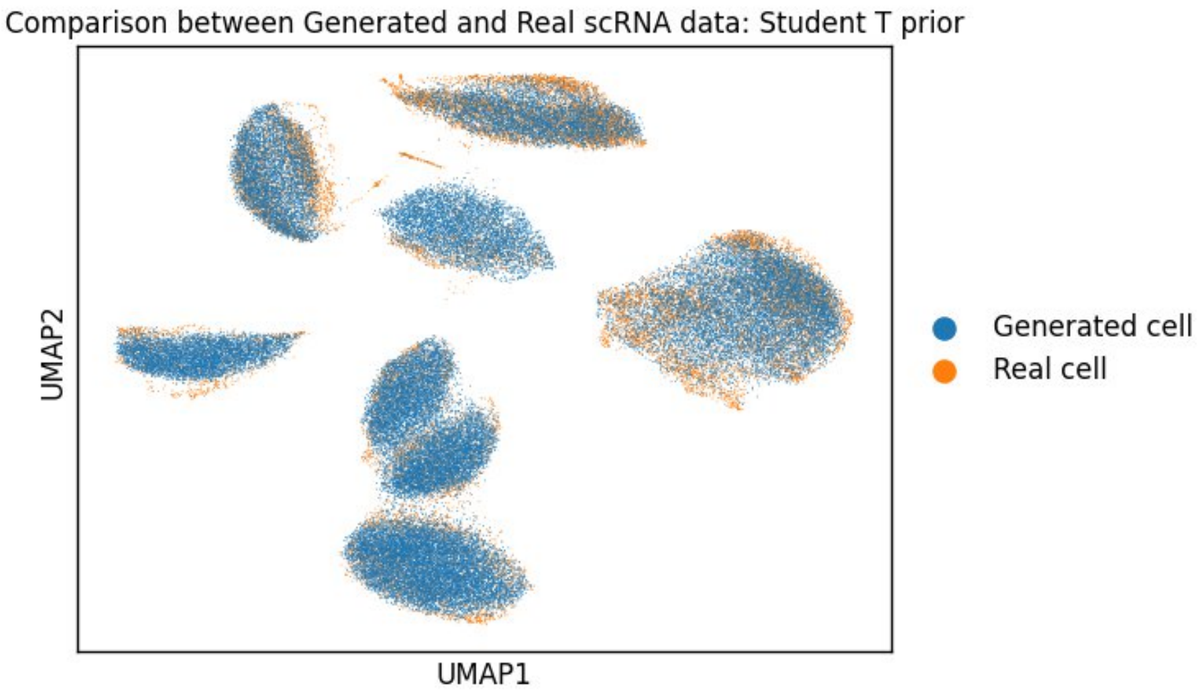

Figure 8. UMAP of real data and generated data (Student's t prior)

Non-statistical Metrics

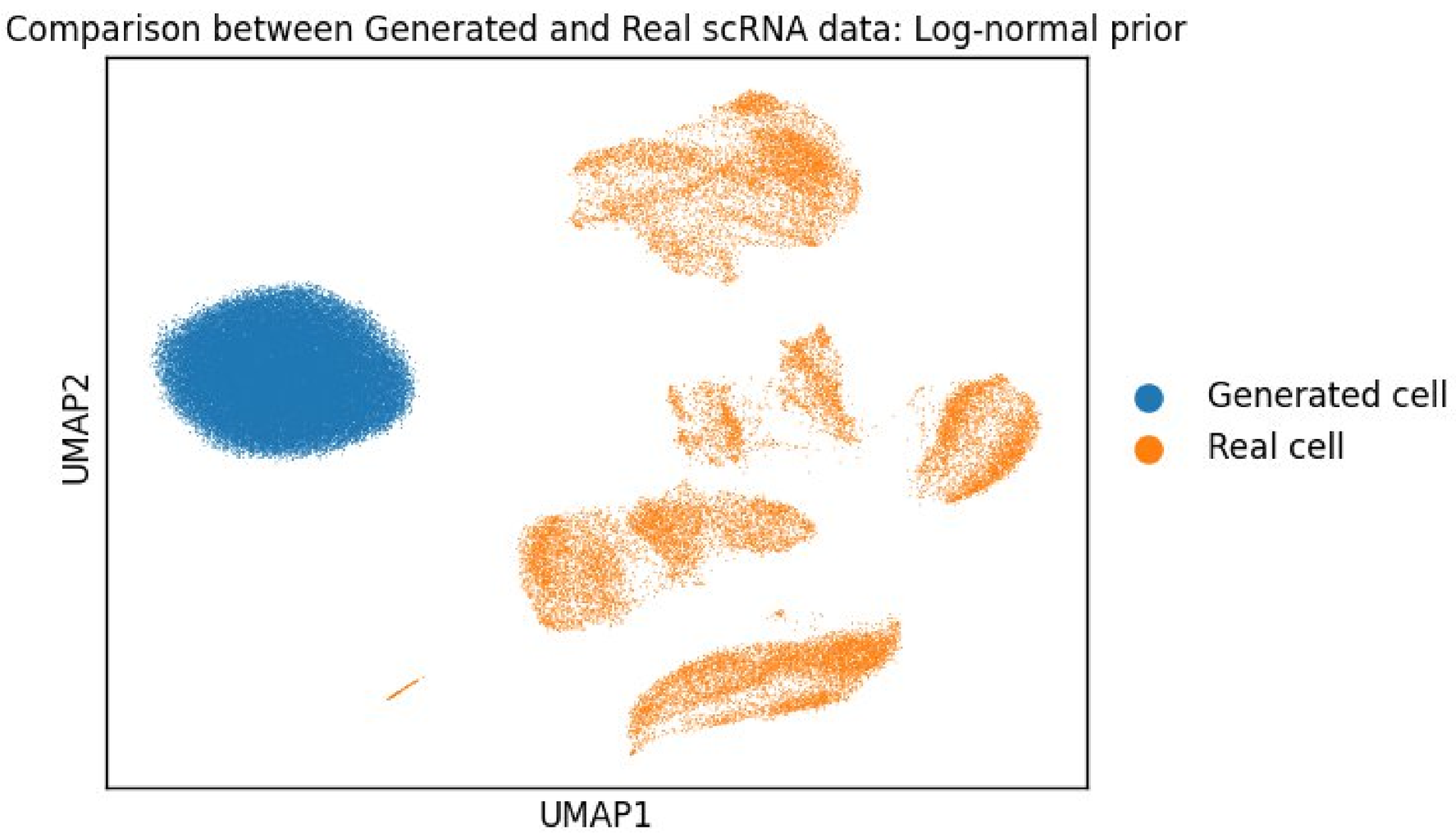

Figure 9. UMAP of real data and generated data (Log-normal prior)

Non-statistical Metrics

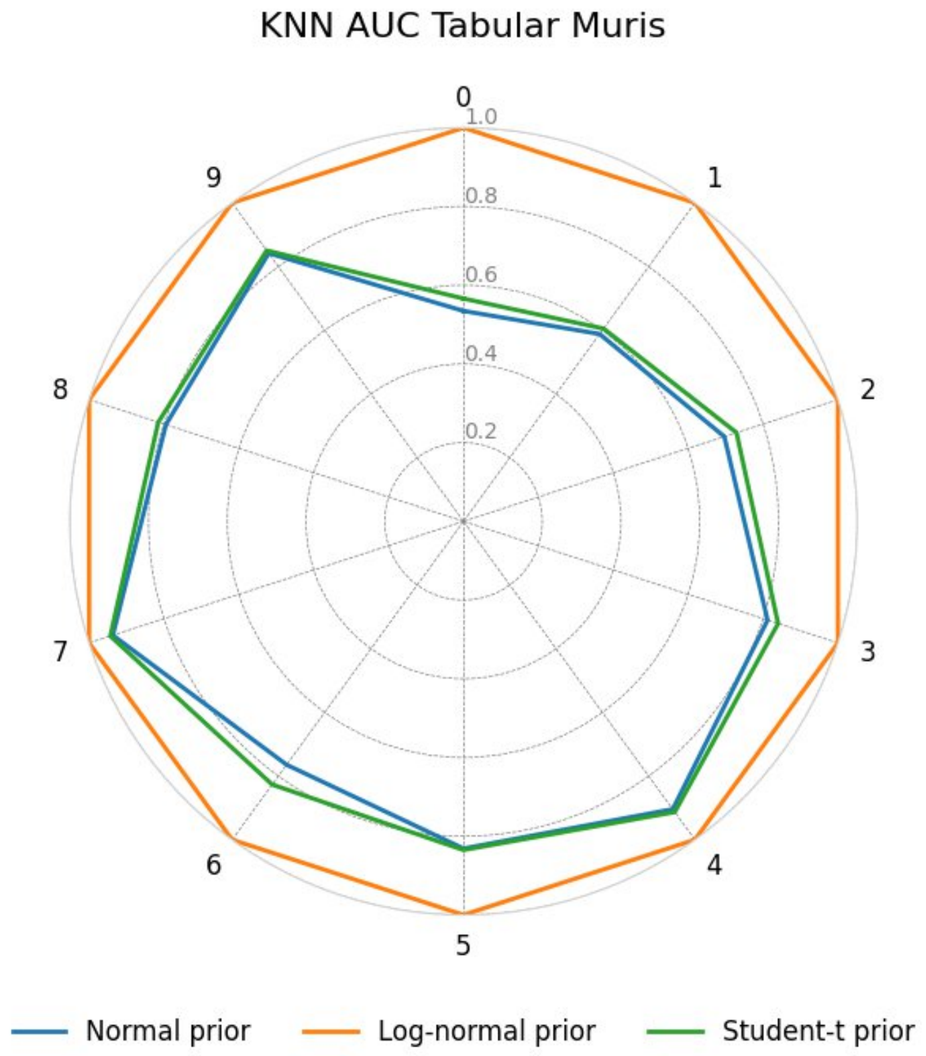

Figure 10. KNN-AUC scores for different prior setting

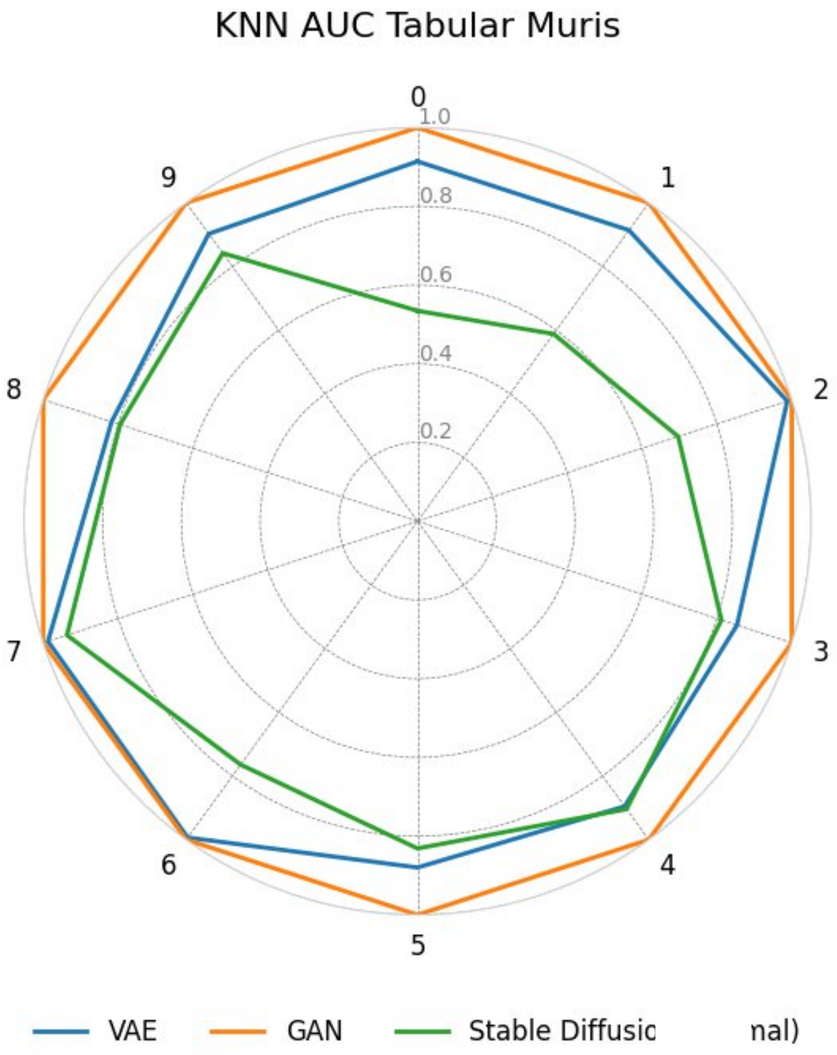

Figure 11. KNN-AUC scores for different models

4.Application

Application of genCell for Lung Fibrosis disease data generation

Application in Lung Fibrosis data generation

- **Statistics** [7]
  - ~875,000 deaths worldwide (2001–2022)
  - 87.5% of deaths occur in patients ≥65 years.
  - Median survival 2–5 years after diagnosis
- **scRNA-seq data**: enables single-cell resolution to uncover [8]
  - cell-based or gene therapies
  - track disease progression
  - identify novel therapeutic targets for precision medicine treatment
- **Objective**: applied genCell for Lung Fibrosis scRNA-seq data generation

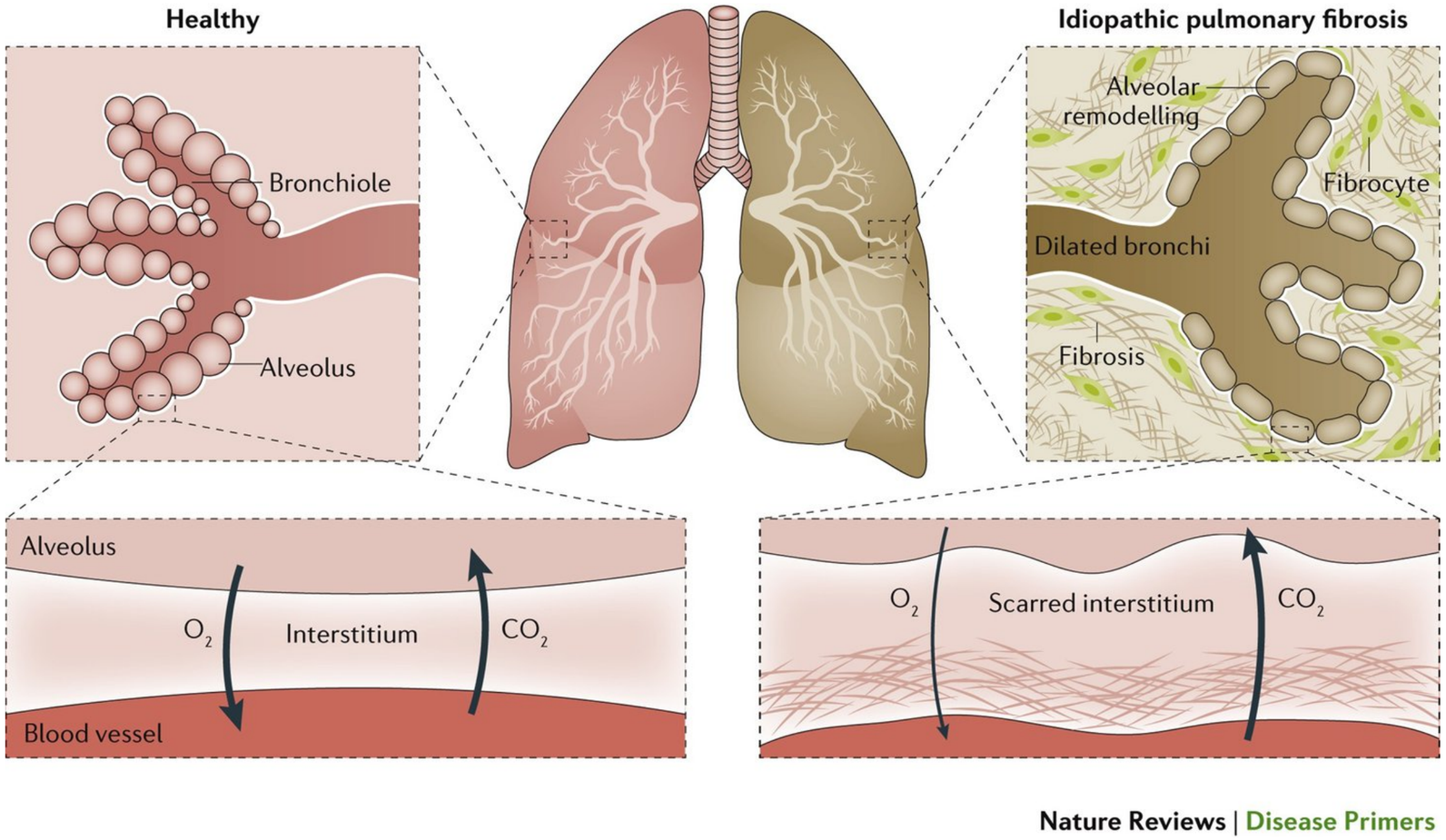

Figure 12. Lung Fibrosis is a progressive lung disease characterized by scarring of lung tissue, leading to stiffness and reduced oxygen exchange [7].

Application in Lung Fibrosis data generation

- **Data source and selection**
  - Source: The Human Lung Cell Atlas (HLCA) [9]
  - 36791 samples (cells) and 1500 features (genes).

- **Data preparation:** the data is divided into 80-20% for train and evaluate sets.

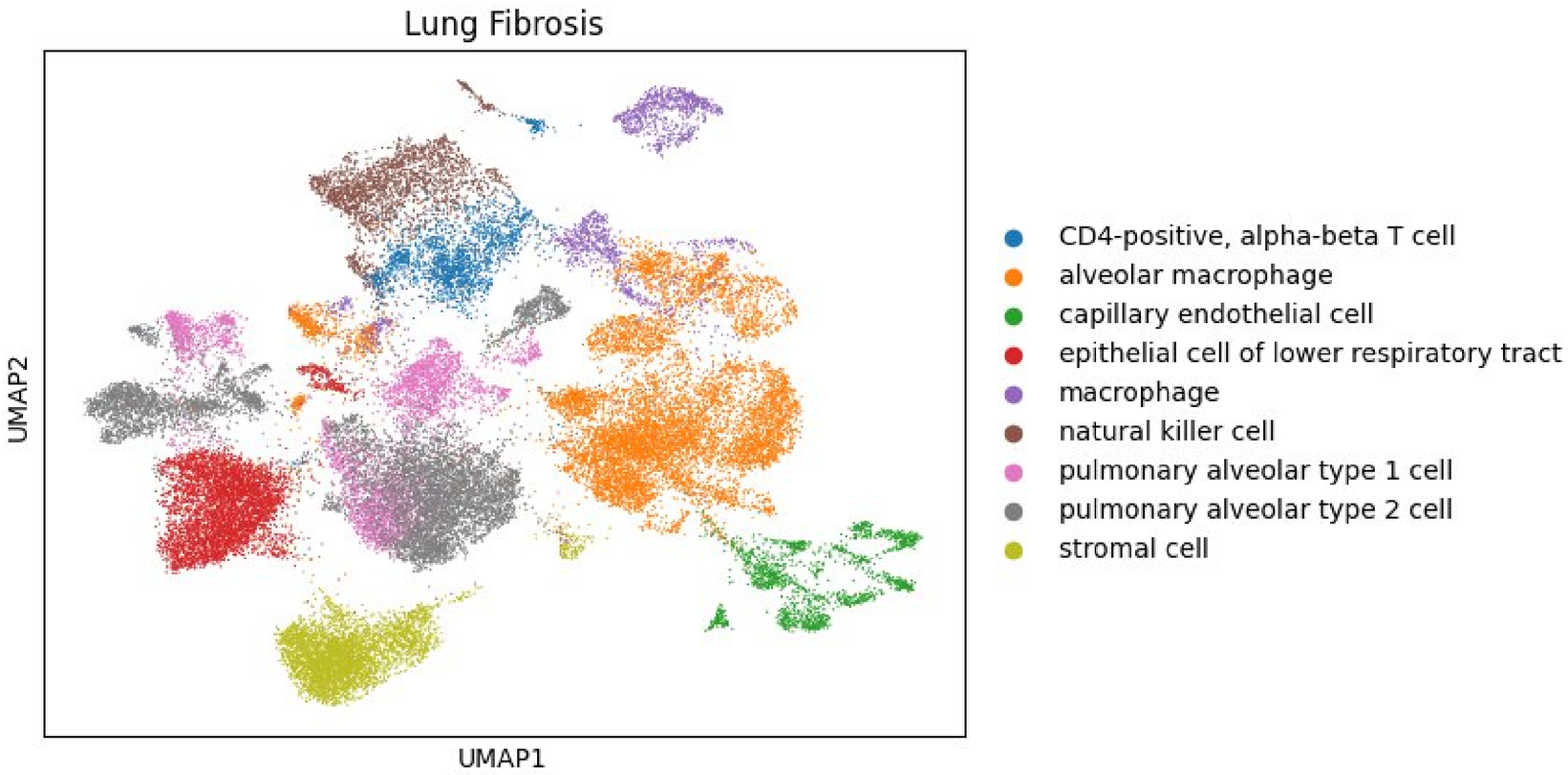

Figure 13. Lung Fibrosis. UMAP visualization for 10 most common cell types

Application in Lung Fibrosis data generation

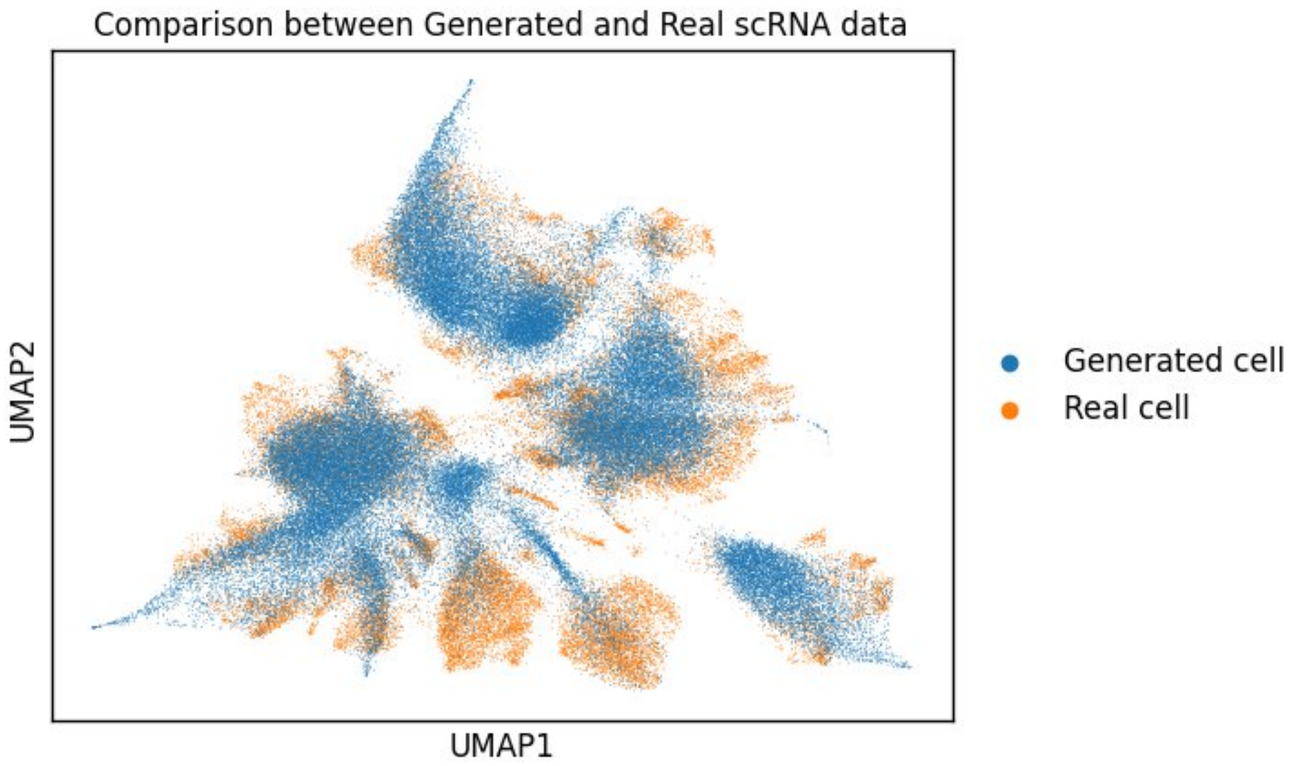

Figure 14. UMAP visualization of the Lung Fibrosis dataset generated by genCell

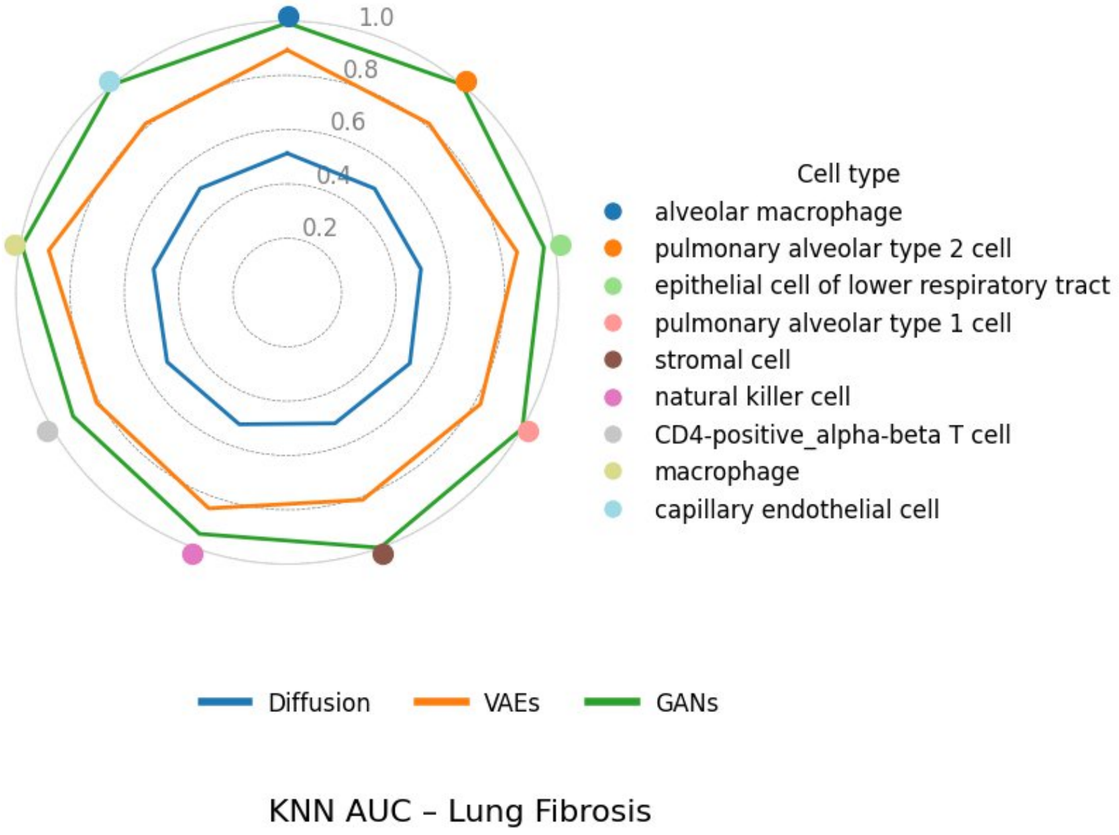

Figure 15. The KNN-AUC score of different cell types in the Lung Fibrosis dataset

Application in Lung Fibrosis data generation

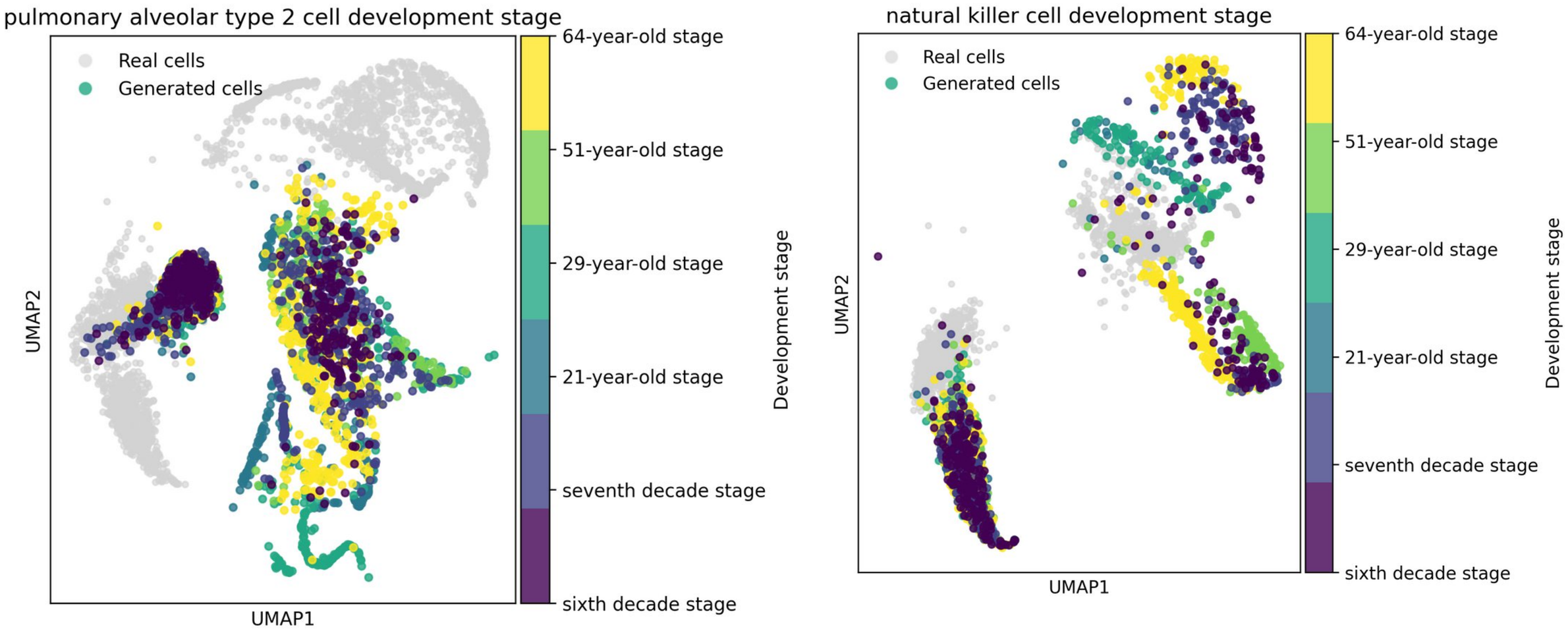

Figure 16. UMAP visualization of different cell types in many development stages

5.Conclusion & Future Improvement

Application in Lung Fibrosis data generation

- Built genCell - a conditional diffusion model with the ability to generate high quality scRNA-seq data and provided a new guidance for future generative model design in single-cell research
- Applied the model effectively for Lung Fibrosis dataset
- More effort in the evaluation metric is needed since the current evaluation metrics are dependent on the known data.

Thank you!

Reference

[1] George Bawa et al. “Single-Cell RNA Sequencing for Plant Research: Insights and Possible Benefits”. In: International Journal of Molecular Sciences 23.9 (2022). ISSN: 1422-0067. URL: https://www.mdpi.com/1422-0067/23/9/4497.
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[3] Haotian Cui et al. “scGPT: toward building a foundation model for single-cell multi-omics using generative AI”. en. In: Nat. Methods 21.8 (Aug. 2024), pp. 1470–1480.
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[7] Ko Harada et al. “Global trends in idiopathic pulmonary fibrosis mortality rates during 2001–2022”. en. In: ERJ Open Res. (June 2025), pp. 00362–02025.
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[9] Lisa Sikkema et al. “An integrated cell atlas of the lung in health and disease”. en. In: Nat. Med. 29.6 (June 2023), pp. 1563–1577.

Code and Dataset availability: https://github.com/chisquare09/genCell